

Recommendations — chondrosarcoma-mets-acetabular-fx-id h-unknown-y34r

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Recommendations — chondrosarcoma-mets-acetabular-fx-idh-un

Profile snapshot

- **Primary site:** bone - left acetabulum (pelvis)
- **Histology:** chondrosarcoma, high-grade with dedifferentiated features (user-stated; consistent with dedifferentiated chondrosarcoma, though a formal pathology report has not been reviewed). Expert bone-sarcoma pathology review to confirm the WHO diagnosis and quantify the dedifferentiated component remains a hardening item, not an open fork.
- **Stage:** IV (metastatic at diagnosis: large left acetabular mass with pathologic acetabular fracture; multiple small pulmonary nodules dispersed through both lungs, consistent with metastases; multiple and dispersed, not oligometastatic). Clinical context: dedifferentiated chondrosarcoma presenting with metastatic disease is aggressive, and the pace of workup and decision-making matters.
- **ECOG:** 2
- **Age band:** 40-49
- **Sex:** female
- **Biomarkers:** IDH1 mutation untested (unknown); IDH2 mutation untested (unknown); Histologic subtype and grade high-grade with dedifferentiated features (user-stated); HEY1-NCOA2 fusion untested (unknown); Comprehensive genomic profile (incl. COL2A1, TERT promoter, CDKN2A/B, TP53) untested (unknown); MSI / dMMR untested (unknown); TMB untested (unknown); PD-L1 untested (unknown)
- **Efficacy/toxicity weight:** 0.8

22 ranked therapeutic options across 4 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Dedifferentiated Histology interventions

1. IACS-6274 (glutaminase-1 inhibitor), phase 1 NCT05039801 - make the screening call now, enroll only on informed choice

Likelihood of effect: Low: no glutaminase inhibitor has a published response in any chondrosarcoma patient; support is grade-linked expression in cell lines (pmid:29576625) with no in-vivo arm.

Toxicity burden: Unknown (phase 1 escalation; no published grade 3+ profile in any chondrosarcoma patient, and adverse-event incidence is the trial's primary endpoint)

Counter-productive MoA: **Moderate** (Glutamine dependence inferred from grade-linked expression in cell lines; no in-vivo chondrosarcoma arm confirms the axis.)

Overall: The one open protocol naming chondrosarcoma as its own treatment-naïve population; what is ranked is the screening call, not a drug with any efficacy estimate.

Key references: NCT05039801 · PMID 29576625

2. Sunitinib plus nivolumab, off-label at first progression

Likelihood of effect: Low: 1 PR among 4 dedifferentiated patients and 5% ORR overall in IMMUNOSARC (pmid:39540661); the microenvironment evidence argues against the axis.

Toxicity burden: High (one treatment-related grade 5 pneumonitis; neutropenia 15%, hypertension 15%, transaminitis 12.5%, 17% discontinued for toxicity)

Counter-productive MoA: **Moderate** (T-cell depletion accelerated growth in the rat model; PD-1 release may favour the dominant macrophage compartment.)

Overall: The only checkpoint dataset in which a dedifferentiated chondrosarcoma responded, bought with a treatment-related grade 5 pneumonitis and a microenvironment that argues against the axis.

Key references: PMID 39540661 · PMID 27312065 · PMID 28344871

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

CD276 (B7-H3) CAR-T on NCT04433221, conditional on a high B7-H3 H-score — Thin evidence

Likelihood of effect: Low: prior is against her, since chondrosarcoma's median B7-H3 H-score of 60 is the lowest of any sarcoma subtype (pmid:42251292) and efficacy is density-dependent.

Toxicity burden: Unknown (no published chondrosarcoma CAR-T safety data; class risks are cytokine release syndrome, neurotoxicity and lymphodepletion cytopenias)

Counter-productive MoA: **Moderate** (Below-threshold B7-H3 density gives sub-lytic CAR engagement and antigen-loss escape rather than tumour control.)

Overall: The antigen is near-universal but the density is not, and the trial pairs the CAR with chemotherapy and a vaccine, so even a response could not be attributed to it.

Key references: NCT04433221 · PMID 30655315 · PMID 42251292

B7-H3-directed antibody-drug conjugates as a class (ifinamab deruxtecan, HS-20093, MGC026, risvutatug rezetecan) — Not enrollable

Likelihood of effect: Unquantified in this disease: sarcoma strata exist in the pan-tumour programmes (pmid:41926962) but no chondrosarcoma efficacy has been reported for any B7-H3 ADC.

Toxicity burden: Unknown for this histology (no chondrosarcoma exposure; the DXd payload class carries interstitial lung disease and cytopenias)

Counter-productive MoA: **Moderate** (Low, heterogeneous B7-H3 density limits payload delivery; bystander killing is the only mechanism that partly offsets it.)

Overall: Four separate B7-H3 ADC programmes, none of which accepts a treatment-naïve patient with an assumed ECOG of 2, and none with a chondrosarcoma result.

Key references: PMID 41926962 · PMID 42251292 · PMID 35149548

Ifinatamab deruxtecan (I-DXd, DS-7300a) on IDEate-PanTumor01 (NCT04145622) — _Consolidated_

Likelihood of effect: Consolidated under the B7-H3 ADC class row; no product-specific chondrosarcoma estimate exists.

Toxicity burden: Unknown for this histology (no chondrosarcoma exposure)

Counter-productive MoA: **Moderate** (Same antigen-density ceiling as the class: low B7-H3 expression limits payload delivery.)

Overall: Sarcoma-inclusive and no IHC entry requirement, but refractory disease and ECOG 0-1 close it to a treatment-naïve patient.

Key references: NCT04145622 · PMID 41926962 · PMID 35149548

HS-20093 on ARTEMIS-002 (NCT05830123) and ARTEMIS-103 (NCT06699576) — _Consolidated_

Likelihood of effect: Consolidated under the B7-H3 ADC class row; no product-specific chondrosarcoma estimate exists.

Toxicity burden: Unknown for this histology (no chondrosarcoma exposure)

Counter-productive MoA: **Moderate** (Same antigen-density ceiling as the class: low B7-H3 expression limits payload delivery.)

Overall: The most sarcoma-permissive B7-H3 protocol running, and the only untreated-disease arm in the class, with chondrosarcoma named in neither cohort list.

Key references: NCT05830123 · NCT06699576 · PMID 42251292

MGC026 (B7-H3-directed ADC), phase 1/1b (NCT06242470) — _Consolidated_

Likelihood of effect: Consolidated under the B7-H3 ADC class row; no product-specific chondrosarcoma estimate exists.

Toxicity burden: Unknown for this histology (no chondrosarcoma exposure)

Counter-productive MoA: **Moderate** (Same antigen-density ceiling as the class: low B7-H3 expression limits payload delivery.)

Overall: The one B7-H3 protocol with no numeric fitness bar on the record, so the eligibility question has to be asked of the site directly.

Key references: NCT06242470 · PMID 42251292

Risvutatumab deruxtecan (GSK5764227) on EMBOLD (NCT06551142) — _Consolidated_

Likelihood of effect: Consolidated under the B7-H3 ADC class row; no product-specific chondrosarcoma estimate exists.

Toxicity burden: Unknown for this histology (no chondrosarcoma exposure)

Counter-productive MoA: Moderate (Same antigen-density ceiling as the class: low B7-H3 expression limits payload delivery.)

Overall: Carried so the reader sees the B7-H3 pipeline is not a single-sponsor bet; entry sits well behind first-line therapy.

Key references: NCT06551142 · PMID 42251292

PD-1 / PD-L1 checkpoint blockade as a class (nivolumab, envafolelimab and others) — _Consolidated_

Likelihood of effect: Consolidated under the pembrolizumab row; the class carries no separate chondrosarcoma efficacy estimate.

Toxicity burden: Low (class profile as for pembrolizumab: anaemia, lymphopenia, immune-related pneumonitis and endocrinopathy)

Counter-productive MoA: Moderate (Macrophage-dominant, immune-excluded microenvironment; T-cell depletion accelerated growth in the rat model.)

Overall: The first-line sarcoma chemo-immunotherapy protocols all carve out cartilage histology; the label-carrying version of this class is the ranked pembrolizumab row.

Key references: PMID 27312065 · PMID 33912442 · PMID 31956474

Nivolumab plus ipilimumab (Alliance A091401 experience) — _Thin evidence_

Likelihood of effect: Not estimable in this histology; excluded upstream with no chondrosarcoma subgroup readout carried into the dossier.

Toxicity burden: Unknown for this histology (no chondrosarcoma subgroup safety readout in the dossier)

Counter-productive MoA: Moderate (Same macrophage-dominant resistance argument as single-agent PD-1 blockade, with a larger immune-toxicity bill.)

Overall: Considered and excluded upstream; recorded so the dual-checkpoint question is visibly asked rather than silently dropped.

Key references: PMID 29370992

Mifamurtide (MEPACT) with doxorubicin, macrophage-directed therapy — _Thin evidence_

Likelihood of effect: Unquantified: one xenograft combination study (pmid:40960554) and no human chondrosarcoma exposure, against the best mechanistic fit in the dossier.

Toxicity burden: Unknown for this histology (no human chondrosarcoma exposure; evidence is a xenograft combination with doxorubicin)

Counter-productive MoA: Moderate (Broad myeloid activation is untested here; the rat data support CD163+ depletion, not indiscriminate macrophage stimulation.)

Overall: Mechanistically the best fit to this tumour's macrophage-dominant microenvironment, and the row that lost on access rather than on biology.

Key references: PMID 40960554 · PMID 28344871 · PMID 31956474

HIF-2alpha inhibition (belzutifan), off-label — Thin evidence

Likelihood of effect: Unquantified: xenograft knockdown evidence only (pmid:33024108), with no clinical exposure in any chondrosarcoma patient.

Toxicity burden: Unknown for this histology (no chondrosarcoma exposure; evidence is xenograft knockdown)

Counter-productive MoA: Low (No plausible counter-productive vector identified; the limit is absent clinical evidence, not mechanism.)

Overall: A real target with knockdown evidence and no clinical result, surfaced so the reader sees it was considered rather than missed.

Key references: PMID 33024108

Evidence in detail

Sunitinib plus nivolumab, off-label at first progression

Palmerini et al. *Cancer* 2025

- **Disease context:** Advanced progressing bone sarcoma (17 osteosarcoma, 14 chondrosarcoma of which 4 dedifferentiated, 8 Ewing, 1 UPS) (2L+) — 40 treated, 38 evaluable by central review; documented progression required; median age 47.
- **n:** 40
- **Toxicities:** neutropenia (grade ≥ 3): 15.0% (n=40); hypertension (grade ≥ 3): 15.0% (n=40); anaemia (grade ≥ 3): 12.5% (n=40); ALT/AST elevation (grade ≥ 3): 12.5% (n=40); pneumonitis (grade ≥ 3): 2.5% (n=40); discontinuation for toxicity (grade any): 17.0% (n=40)
- **Efficacy:** PFS_rate 42% (95% CI 27-58); PFS 3.8 months (95% CI 2.7-4.8); OS 11.9 months (95% CI 5.6-18.2); ORR 5%

Kostine et al. *Mod Pathol* 2016

- **Disease context:** Chondrosarcoma, all subtypes (tissue microarray + whole-section validation) (any) — TMA: 119 conventional, 19 mesenchymal, 20 clear-cell, 22 dedifferentiated; independent whole-section dedifferentiated cohort n=21.
- **n:** 181
- **Toxicities:** Not applicable (no therapeutic exposure).
- **Efficacy:** biomarker PD-L1 positive 9/22 (41%) dedifferentiated on TMA; 11/21 (52%) in independent whole-section cohort; biomarker PD-L1 0/119 conventional, 0/19 mesenchymal, 0/20 clear-cell

PD-1 / PD-L1 checkpoint blockade as a class (nivolumab, envafolelimab and others)

Kostine et al. *Mod Pathol* 2016

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TMA: 119 conventional, 19 mesenchymal, 20 clear-cell, 22 dedifferentiated; independent whole-section dedifferentiated cohort n=21.

- **n:** 181
- **Toxicities:** Not applicable (no therapeutic exposure).
- **Efficacy:biomarker** PD-L1 positive 9/22 (41%) dedifferentiated on TMA; 11/21 (52%) in independent whole-section cohort; **biomarker** PD-L1 0/119 conventional, 0/19 mesenchymal, 0/20 clear-cell

****Nota et al. *Front Oncol* 2021****

- **Disease context:Conventional and dedifferentiated chondrosarcoma (tissue microarray)** (any) — 52 conventional and 24 dedifferentiated specimens (MGH).
- **n:** 76
- **Toxicities:** Not applicable (no therapeutic exposure).
- **Efficacy:biomarker** CD8+ TILs in 90% of dedifferentiated vs 21% of conventional tumours; **biomarker** B7-H3 expressed in 96% of dedifferentiated vs 69% of conventional tumours; **biomarker** PD-1 53% and PD-L1 33% of dedifferentiated tumours; PD-L1 associated with shorter time to metastasis

Nivolumab plus ipilimumab (Alliance A091401 experience)

****D'Angelo et al. *Lancet Oncol* 2018****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Unmeasured Molecular Landscape interventions

1. Abemaciclib on NCT04040205 (CDK-pathway-altered bone and soft-tissue sarcoma), pre-positioned for the post-anthracycline line

Likelihood of effect: Low to unknown: no published chondrosarcoma efficacy for any CDK4/6 inhibitor; the case rests on preclinical work (pmid:30808351, pmid:18624751) plus a named-histology biomarker-gated seat.

Toxicity burden: Moderate (diarrhoea, neutropenia, venous thromboembolism - class profile; no chondrosarcoma-specific grade 3+ table in the dossier)

Counter-productive MoA:Moderate (CDK4/6 inhibition is cytostatic; Rb-null or CDK-independent clones keep cycling despite target engagement.)

Overall:The cleanest trial design in the screen, dedifferentiated chondrosarcoma named and central Rb IHC pre-specified, sitting behind three untested gates and an ECOG 0-1 bar.

Key references: NCT04040205 · PMID 30808351 · PMID 18624751

2. Pembrolizumab monotherapy, second line, conditional on a TMB-high or MSI-high call

Likelihood of effect: Low: 1 response among 5 chondrosarcoma patients in SARC028 (pmid:28988646); the

on-label TMB-high route is a low-prior bet in a typically TMB-low disease.

Toxicity burden: Low (anaemia 14%, lymphopenia 12%, prolonged aPTT 10%; immune-related pneumonitis, adrenal insufficiency and nephritis reported)

Counter-productive MoA: **Moderate** (Macrophage-dominant microenvironment; T-cell depletion accelerated growth in the rat model, so PD-1 release may not be the lever.)

Overall: Thin but genuine disease-specific signal with the best characterised safety of anything non-cytotoxic here; a TMB-high or MSI-high call moves it on-label and out of this table.

Key references: PMID 28988646 · PMID 32919526 · PMID 37174084

3. TAPUR (NCT02693535): marketed targeted agents assigned by genomic match, pre-positioned for the line after first-line chemotherapy

Likelihood of effect: Unquantified: a basket of single-arm matched cohorts with no chondrosarcoma cohort result. An access decision carrying no efficacy estimate.

Toxicity burden: Unknown (agent assigned by genomic match; toxicity is whatever the matched marketed drug carries)

Counter-productive MoA: **Low** (No mechanism named until a match exists; the risk is second-line time spent on a search that finds nothing.)

Overall: An ECOG 0-2 protocol seat she may clear as written, wholly conditional on a panel that chondrosarcoma genomes often decline to fill.

Key references: NCT02693535

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

NY-ESO-1-directed TCR-engineered autologous T cells (NCT05620693) — _Not enrollable_

Likelihood of effect: Not assessable: both gating tests are unordered and the enrolling condition is soft-tissue sarcoma rather than bone.

Toxicity burden: Unknown (no chondrosarcoma exposure; lymphodepletion and cell-therapy class risks apply)

Counter-productive MoA: **Moderate** (Antigen-loss escape and lymphodepletion in a fast-moving tumour whose manufacturing window competes with disease pace.)

Overall: Unassessable rather than negative: the two gating tests are unordered, and the enrolling condition is soft-tissue sarcoma.

Key references: NCT05620693

Tebentafusp (KIMMTRAK), gp100 peptide-HLA-A*02:01 ImmTAC (NCT06942442) — Not enrollable

Likelihood of effect: None expected: gp100 is melanocytic and is not expressed in cartilage tumours; the protocol enrolls clear cell sarcoma only.

Toxicity burden: Unknown for this histology (no cartilage-tumour exposure)

Counter-productive MoA:Low (Target antigen absent, so the mechanism has no vector in this tumour at all.)

Overall:Carried as the reference point for the HLA-restricted platform and the reason to draw HLA typing, not as an option in this histology.

Key references: NCT06942442

Evidence in detail

Pembrolizumab monotherapy, second line, conditional on a TMB-high or MSI-high call

Tawbi et al. *Lancet Oncol* 2017

- **Disease context:**Advanced bone sarcoma (osteosarcoma 22, Ewing 13, chondrosarcoma 5 evaluable) (2L+) — Up to 3 prior lines; biopsy-accessible lesion required; bone cohort 42 treated, 40 evaluable, including 5 chondrosarcoma without subtype restriction.
- **n:** 84
- **Toxicities:**anaemia (grade ≥ 3): 14.0% (n=42); lymphocyte count decreased (grade ≥ 3): 12.0% (n=42); activated partial thromboplastin time prolonged (grade ≥ 3): 10.0% (n=42); platelet count decreased (grade ≥ 3): 7.0% (n=42); treatment-emergent serious AE (grade any): 12.0% (n=42); pneumonitis (immune-related SAE) (grade any): 2.4% (n=84)
- **Efficacy:**ORR 20%; ORR 5%

Bui et al. *Cancers (Basel)* 2023

- **Disease context:**Advanced dedifferentiated chondrosarcoma (2L+) — ICI-treated subset of the 74-patient DDCS cohort; the pembrolizumab responder was PD-L1 TPS 7% / CPS 8, microsatellite-stable.
- **n:** 11
- **Toxicities:** No per-term AE reporting (retrospective).
- **Efficacy:**ORR 1 PR / 5 pembrolizumab-treated; other prolonged SD 24 and 48 weeks

Idh Mutation interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Ivosidenib (AG-120), off-label mutant-IDH1 inhibition — Thin evidence

Likelihood of effect: Low in her subtype: the 23.1% ORR belongs to 13 conventional-histology patients (pmid:40100120); the only dedifferentiated figure is olutasidenib at mPFS 1.5 months.

Toxicity burden: Moderate (all-cause grade 3+ TEAEs 57% of 21, only one treatment-related: hypophosphatemia; grade 1-2 QT prolongation in 5)

Counter-productive MoA: **Moderate** (Tumour appears to have outgrown IDH dependence: D-2-HG fell over 90% with no phenotypic change to 20 passages.)

Overall: A positive IDH1 call opens an off-label conversation rather than the targeted rescue the word mutation suggests, and both chondrosarcoma IDH trials exclude her by name.

Key references: PMID 32208957 · PMID 40100120 · PMID 41895355

Enasidenib (AG-221), mutant-IDH2 inhibition via NCT06176989 — _Not enrollable_

Likelihood of effect: Unquantified: single-arm, single-site, no results reported; her acetabular primary sits outside the enrolling anatomy regardless of IDH2 status.

Toxicity burden: Unknown for this histology (no chondrosarcoma safety readout exists for this protocol)

Counter-productive MoA: **Moderate** (Same differentiation-block rationale as IDH1 inhibition, and the same evidence that dedifferentiated tumours have escaped it.)

Overall: The only active IDH2-directed protocol taking chondrosarcoma, closed to her by an anatomical criterion no assay result can change.

Key references: NCT06176989

LY3410738 (covalent mutant IDH1/IDH2 inhibitor), NCT04521686 — _Unavailable_

Likelihood of effect: Not assessable: the programme closed to accrual with no published efficacy readout.

Toxicity burden: Unknown (dose-finding programme closed; no safety readout published)

Counter-productive MoA: **Moderate** (Same IDH-dependence question as ivosidenib, unresolved because no readout was published.)

Overall: Would have been the best-fitting IDH protocol in the screen, with no conventional-histology restriction; accrual closed in 2023.

Key references: NCT04521686

Olaparib plus ceralasertib (AZD6738) in IDH-mutant solid tumours, NCT03878095 — _Not enrollable_

Likelihood of effect: Low: the chondrosarcoma-specific preclinical test of this premise was negative (pmid:31810230), and the trial is closed to accrual.

Toxicity burden: Unknown for this histology (NCT03878095 results unpublished; no chondrosarcoma safety readout)

Counter-productive MoA: Moderate (Talazoparib sensitivity did not sort by IDH status and D-2-HG normalisation produced no synthetic lethality.)

Overall: The synthetic-lethality premise was tested in chondrosarcoma lines and failed; the trial is closed to accrual besides.

Key references: NCT03878095 · PMID 31810230

Evidence in detail

Ivosidenib (AG-120), off-label mutant-IDH1 inhibition

Tap et al. *J Clin Oncol* 2020

- **Disease context:** Advanced mutant-IDH1 chondrosarcoma (any) — 21 patients (12 escalation, 9 expansion); 11 had prior systemic therapy; histology includes conventional and non-conventional chondrosarcoma. IDH1 mutation required.
- **n:** 21
- **Toxicities:** any TEAE (grade ≥ 3): 57.1% (n=21); hypophosphatemia (treatment-related) (grade ≥ 3): 4.8% (n=21); fatigue (grade any): 42.9% (n=21); diarrhea (grade any): 33.3% (n=21); nausea (grade any): 28.6% (n=21); QT prolongation (grade 1-2): 23.8% (n=21)
- **Efficacy:** PFS 5.6 months (95% CI 1.9-7.4); PFS_rate 39.5%; DCR 52%; biomarker plasma 2-HG reduced 14-94.2% to healthy-range levels

Tap et al. *Clin Cancer Res* 2025

- **Disease context:** Advanced mutant-IDH1 conventional chondrosarcoma (any) — 21 chondrosarcoma patients dosed in the phase 1; efficacy reported only for the 13 with conventional histology. Database lock March 2024.
- **n:** 13
- **Toxicities:** any treatment-related AE (grade any): 71.4% (n=21); any serious AE (grade any): 28.6% (n=21)
- **Efficacy:** ORR 23.1%; DoR 53.5 months; PFS 7.4 months; other 46.2% >1 yr, 30.8% >6 yr, 23.1% >7 yr on treatment

Remiszewski et al. *Crit Rev Oncol Hematol* 2026

- **Disease context:** Dedifferentiated chondrosarcoma, neoadjuvant to palliative (any) — Narrative synthesis of the dedifferentiated-specific literature; metastases in ~40-80%, contemporary 5-year OS ~7-24%.
- **n:** —
- **Toxicities:** Review notes lower pathological response rates than osteosarcoma and substantial haematological toxicity with osteosarcoma-type regimens; no primary AE table.
- **Efficacy:** ORR ~20%; PFS 4-5 months; OS_rate 7-24%; PFS 1.5 months

Acetabular Lesion Pathologic Fracture interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable

(program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Proton beam or carbon-ion radiotherapy to the acetabular lesion (NCT02838602, NCT05033288) **— _Not enrollable_**

Likelihood of effect: Low for her situation: the local-control figures come from skull-base, non-metastatic, curative-intent cohorts (pmid:40889733), and both particle protocols exclude distant metastases.

Toxicity burden: High (neurological deficit 52%, pelvic insufficiency fracture 48%, femoral head necrosis 37% among periacetabular survivors)

Counter-productive MoA:High (Irradiating a fractured, unfixed pelvis produced atrophic non-union in the rat model; local control bought at structural cost.)

Overall:The one option all five personas declined independently: the toxicity ledger lands hardest in exactly her anatomy, and both protocols exclude metastatic disease.

Key references: PMID 32658315 · PMID 26150259 · NCT02838602

Evidence in detail

Proton beam or carbon-ion radiotherapy to the acetabular lesion (NCT02838602, NCT05033288)

Takenaka et al. *Cancer* 2020

- **Disease context:**Unresectable pelvic bone sarcoma (long-term survivors) (1L) — 29 patients without local recurrence or metastasis, drawn from 112 CIRT-treated; mean follow-up 93 months; periacetabular tumours analysed as a risk factor.
- **n:** 29
- **Toxicities:**femoral head necrosis (grade any): 37.0% (n=29); pelvic insufficiency fracture (grade any): 48.0% (n=29); neurological deficit (grade any): 52.0% (n=29)
- **Efficacy:**AE_rate 37%; AE_rate 48%; AE_rate 52%; other mean MSTS 53%, TESS 64%, EQ-5D 0.587

Outani et al. *Int J Clin Oncol* 2016

- **Disease context:**Pelvic chondrosarcoma treated with curative intent (1L) — 31 patients, 1983-2014, non-metastatic; median age 43; median follow-up 66 months.
- **n:** 31
- **Toxicities:** No per-term complication table in the source abstract; functional outcome favoured CIRT.
- **Efficacy:**OS_rate 72%; other OS not different between surgery and CIRT; other local recurrence 32% overall; other function better after CIRT than surgery (mean MSTS 59% overall)